

● EASY

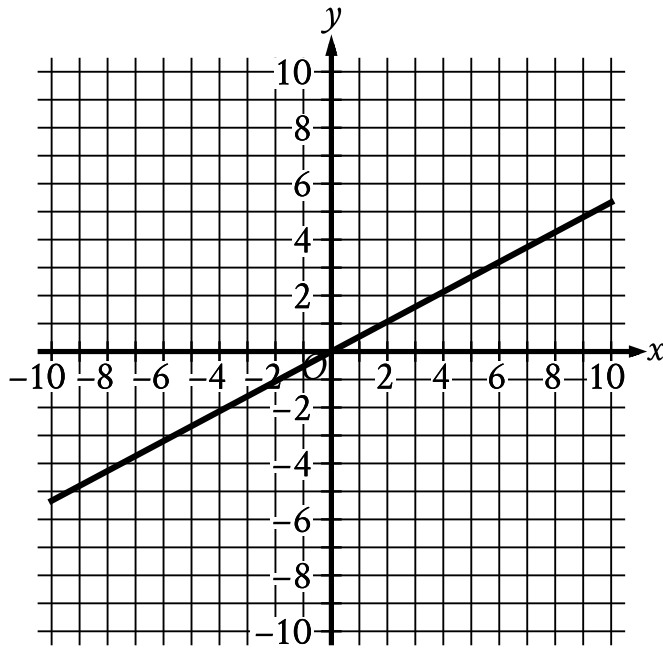
● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

11

10 questions · ~15 min

The graph of function f is shown, where $y = fx$.



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (negative 6 comma negative $\frac{16}{5}$)
 - (0 comma 0)
 - (6 comma $\frac{16}{5}$)

Which of the following describes function f ?

- A. Increasing linear
- B. Decreasing linear
- C. Increasing exponential

D. Decreasing exponential

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio of t to u is 1 to 2, and $t = 10$. What is the value of u ?

- A. 2
- B. 5
- C. 10
- D. 20

There are a total of 840 seats in a school auditorium. During an assembly, students occupied 50% of the seats in the auditorium. How many seats did the students occupy during this assembly?

- A. 25
- B. 50
- C. 420
- D. 790

On a street with 7 houses, 2 houses are blue. If a house from this street is selected at random, what is the probability of selecting a house that is blue?

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{5}{7}$

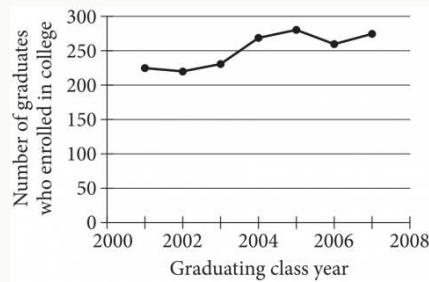
D. $\frac{7}{7}$

Five *Eretmochelys imbricata*, a type of sea turtle, each have a nest. The table shows an original data set of the number of eggs that each turtle laid in its nest.

Nest	Number of eggs
A	149
B	144
C	148
D	136
E	139

A sixth nest with 121 eggs is added to create a new data set. Which of the following correctly compares the means of the two data sets?

- A. The mean of the original data set is greater than the mean of the new data set.
- B. The mean of the original data set is less than the mean of the new data set.
- C. The means of both data sets are equal.
- D. There is not enough information to compare the means.



The line graph shows the number of graduates from the classes of 2001 through 2007 at a certain school who enrolled in college within 24 months of graduation. Of the following, which class had the fewest graduates who enrolled in college within 24 months of graduation?

- A. 2002
- B. 2004
- C. 2005
- D. 2007

x	y
1	4
3	12
5	20
40	k

In the table above, the ratio of y to x for each ordered pair is constant. What is the value of k ?

- A. 28
- B. 36
- C. 80
- D. 160

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

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Of the 8 planets in our solar system, 4 are considered rocky. If a student randomly selects 1 of those 8 planets as a topic for a report, what is the probability that the selected planet will be rocky?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. 2

